

E-Waste Robotic Battery Extraction Project

Robotic Arm Evaluation Report

Sprint 1 - T1 2026

Robotics Sub-Team | Focus Device: Samsung Smartphones

1. Introduction

The E-Waste Robotic Battery Extraction project aims to develop a safe, efficient, and cost-effective robotic system capable of autonomously extracting lithium-ion batteries from end-of-life electronic devices. This project will solely focus on Samsung devices for the T1 the focus can be later increased to electronic appliances and other smartphones. This focused scope allows the team to establish a solid technical foundation for validated robot configuration, extraction method, and coordinate detection approach.

This report evaluates three candidate robotic arm configurations for the project: the SCARA (Selective Compliance Assembly Robot Arm), the Delta Robot, and the 6-Axis Articulated Arm. The evaluation draws on findings from two key research papers reviewed by the team, as well as technical specifications relevant to our project constraints.

The report is structured as follows:

- Section 2 defines the evaluation criteria.
- Section 3 provides detailed analysis of each configuration.
- Section 4 presents a comparative table.
- Section 5 discusses pros, cons, and key challenges for 6-axis robot.

2. Evaluation Criteria

The three configurations are evaluated across eight criteria, selected based on the project's four success pillars (Safety, Sustainability, Affordability, Efficiency) and the specific demands of disassembling Samsung smartphones.

Criterion	Description
Degrees of Freedom (DoF)	Number of independent movement axes; higher DoF which means greater reach and adaptability

Positional Accuracy	Repeatability and precision in XYZ space, critical for targeting small battery connectors
Payload Capacity	Maximum weight the arm can reliably handle during operation
Speed	Cycle time for pick-and-place or extraction movements
Budget Compatibility	Commercial availability and cost relative to our \$200/trimester budget
ROS/Software Integration	Compatibility with ROS and Python-based control pipelines
Safety in Battery Handling	Risk of puncture, short circuit, or excessive force during extraction
Workspace Suitability	Whether the arm's workspace geometry suits a flat, compact device like a Samsung phone

3. Robotic Arm Configurations — Detailed Analysis

3.1 SCARA Robot (Selective Compliance Assembly Robot Arm)

A SCARA robot has two horizontal revolute joints (typically 4 DoF including a vertical Z-axis and wrist rotation) that give it excellent compliance in the horizontal XY plane while maintaining rigidity vertically. This architecture is well-suited for repetitive, planar tasks where precise lateral positioning is required, such as screw removal (in later cohorts) and battery connector detachment in a flat-lying device like a Samsung phone.

SCARA arms are widely used in electronics assembly due to their speed and precision at moderate costs. Research by Kranz et al. (2023) demonstrated a UR5e a 6-axis arm for e-waste manipulation, but noted that much of the required manipulation in flat electronic devices occurs in a predominantly horizontal workspace, which SCARA naturally accommodates. From a Samsung phone extraction standpoint, the device lies flat on a fixture, and the primary actions all occur in this horizontal plane.



Figure 1: SCARA Robot (<https://automatedsolutions.com.au/fanuc-scara-robots/>)

3.2 Delta Robot

A Delta robot consists of three arms connected to a central end-effector platform via parallelogram linkages. It is optimised for very high-speed pick-and-place operations within a dome-shaped workspace. Delta robots excel in speed (often 150+ picks per minute in industrial settings) but have limited payload capacity (typically under 2 kg) and restricted ability to apply rotational force or torque.

In the context of e-waste battery extraction, the Delta robot's high speed is largely unnecessary. Extraction is a careful, sequential process rather than a rapid pick-and-place task. More critically, the Delta robot lacks the torque and angular dexterity required to apply controlled lateral prying force. Ude et al. (2025) found that battery removal from electronic devices demands 'advanced tactile motor skills, precise manipulation, and the ability to apply the correct amount of force', capabilities that are not naturally suited to the Delta configuration.



Figure 2: Delta Robot (<https://www.weiss-world.com/us-en/products/robots-10327/delta-robots-211>)

3.3 6-Axis Articulated Arm

A 6-axis articulated arm (such as the UR5e used in Kranz et al., 2023, or the modular arms used in Ude et al., 2025) provides full rotational freedom across six joints, enabling the end-effector to reach virtually any position and orientation within its workspace. This makes it the most versatile configuration for complex disassembly tasks, for prying in confined spaces, and fine battery connector manipulation.

The 6-axis arm was specifically used in both reference papers for e-waste manipulation. Ude et al. (2025) used dual 6-axis arms with soft end-effectors to handle diverse device types, demonstrating successful battery removal with adaptive force-impedance control. Kranz et al. (2023) used the UR5e 6-axis arm with a Robotic Hand-E gripper running ROS Noetic, achieving task board localisation errors of just 1.4 ± 0.66 mm. However, industrial 6-axis arms (UR, KUKA, ABB) are expensive, often far exceeding the project budget for outright purchase, though budget-friendly alternatives exist (e.g., low-cost 6-DoF desktop arms such as myCobot or Elephant Robotics).



Figure 3: 6-axis Robot (<https://www.robomotion.co.uk/the-many-uses-for-industrial-6-axis-robots/>)

4. Comparative Analysis Table

The following table scores each configuration across the defined criteria. Ratings are qualitative: High / Medium / Low, with brief justification.

Criterion	SCARA	Delta Robot	6-Axis Arm
Degrees of Freedom	4 DoF is sufficient for planar tasks	3–4 DoF is limited to vertical dome workspace	6 DoF means full orientation control
Positional Accuracy	High = ± 0.01 mm typical in XY plane	High = ± 0.1 mm but limited axes	High = ± 0.03 – 0.1 mm (model dependent)
Payload Capacity	Medium = 1–10 kg typical	Low = typically under 2 kg	Medium-High = 3–10 kg typical
Speed	High = fast horizontal movements	Very High = fastest pick-and-place	Medium = deliberate, controlled motion
ROS Integration	Medium = ROS drivers available but less common than 6-axis	Low = limited ROS ecosystem support	High = UR5e, myCobot, and others have mature ROS support
Safety in Battery Handling	Medium-High = rigid vertical axis prevents unintended Z force	Low-Medium = limited force control, risk of over-speed	High = force-impedance control possible, compliant end-effectors supported
Workspace Suitability (Samsung Phone)	High = flat XY plane matches phone disassembly	Low = dome workspace unsuitable for flat device work	High = full orientation enables all angles of approach
Simulation Feasibility	Medium = limited simulation environments	Low = niche simulation support	High = Gazebo, RViz, MoveIt! all support 6-axis arms

Overall Rating for This Project	RECOMMENDED — primary candidate	NOT RECOMMENDED — poor fit	STRONG CANDIDATE — especially for simulation phase
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5. Pros, Cons, and Key Challenges

6-Axis Articulated Arm

Pros

- Full 6 DoF can approach the Samsung phone's battery from any angle, essential for connector detachment and adhesive prying at varied orientations.
- Force-impedance control (UFIC as used in Ude et al., 2025) enables the arm to apply controlled, safe forces during extraction and is critical for avoiding battery damage.
- Best-in-class ROS/MoveIt! ecosystem the UR5e used in Kranz et al. (2023) ran on ROS Noetic with full motion planning support.
- Compatible with soft grippers and adaptive end-effectors as researched in Ude et al. (2025).
- Supports simulation in Gazebo and RViz which aligns with the project's simulation-first approach for this trimester.
- Budget desktop 6-axis arms (myCobot 280, Elephant Robotics) available at accessible price points.

Cons

- More complex kinematics and inverse kinematics and motion planning require more computational setup.
- Industrial-grade units (UR5e, KUKA) are far beyond the project's \$200 trimester budget for purchase.
- More moving parts increase maintenance complexity.
- Execution speed is lower than SCARA or Delta which is acceptable for this project but worth noting.

Key Challenges

- Budget constraint is the primary challenge if physical hardware is required in Sprint 3, a low-cost 6-axis desktop arm must be identified, verified for ROS compatibility, and procured.
 - Complex motion planning for confined-space operations (battery compartment access) requires significant software development effort.
 - As noted by Ude et al. (2025), motion adaptation for specific device geometries 'can require reparameterization for certain designs' Samsung phones vary across generations and models.
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8. References

Ude, A., Simonic, M., Kuster, B., Mavsar, M., Bem, M., Ruiz, S., ... & Worgotter, F. (2025). The Challenges of Using Robots to Automate the Recycling of Electronic Devices: A New Approach Incorporating AI, Modularity, and Soft Robotics. IEEE Robotics & Automation Magazine. DOI: 10.1109/MRA.2025.3619331

Kranz, P., Ali, U., Mueller, A., Hornauer, M., Loeser, M., Sukkar, F., Willert, V., & Kaupp, T. (2023). Towards Recycling E-Waste Using Vision and Robotic Manipulation. Technical University of Applied Sciences Wuerzburg-Schweinfurt & Australian Cobotics Centre, University of Technology Sydney.